

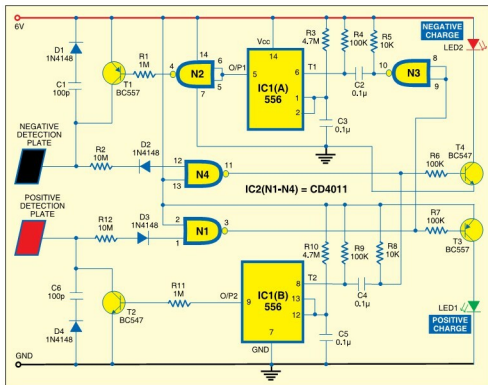


Fig. 1: Circuit of electronic electroscope

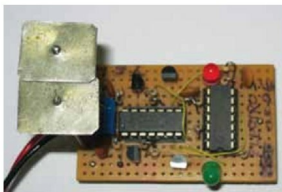


Fig. 2: Author's prototype

and the input voltage of gate N1 goes high. This makes the gate output low and transistor T3 conducts to light up the green LED.

The process is similar during negative charge detection. Here transistor T4 conducts to light up the red LED.

When a negatively charged object is brought near the detecting plate, it also affects the positive-detecting plate. The negatively charged object pushes the electrons in the positive-detecting plate towards its input capacitor and the plate becomes negatively

(V+0.6), the gate will not conduct heavily to return the electrons immediately. In this case, the charge developed across the capacitor takes a long time to decay and the gate becomes locked.

To overcome this problem, two monostable multivibrators built around IC1(A) and IC1(B) have been incorporated for each arm. The high output of monostable IC1(B) shorts capacitor C6 to the ground for a specific time as soon as the output of the negative-sensing gate (N4) goes from high to low. Similarly, the high output

of IC1(A) charges capacitor C1 for a specific time as soon as the output of the positive-sensing gate (N1) goes from low to high.

A similar arrangement is provided for both the positive- and negative-detecting sides. So when a positively charged object is withdrawn, the negative-charge-detecting input capacitor is shorted. The monostables prevent locking of any of the two gates and allow the scope to function smoothly.

The electroscope can be designed for bench-mounting or more compactly inside a gluestick tube with identical metallic detector plates on the cap. Two CR2032 (3V) cells are used to power the circuit. Take the tube slowly towards the charged object or bring the object slowly towards

the scope. Either of the two LEDs will glow depending on the polarity of the charge on the object. The author's prototype is shown in Fig. 2.

To test the circuit after construction, whack your hair with a comb and bring the comb near the scope slowly. The red LED should glow to indicate a negative charge. This glow should be constant as long as the comb is near the detecting plates.

It is also possible to use the scope as a direct charge detector within limits by bringing the corresponding plates (or just the lead) in direct contact with the point of charge accumulation in a circuit. The sensitivity can be increased to some extent by decreasing the values of capacitors C1 and C6.

EFY note. External reset can be provided using an SPDT switch (not shown in the circuit) between the output of the monostable and the 1-mega-ohm resistor, with its pole connected to the 1-mega-ohm end. Connect one throw to the monostable and the other to +V or 0, as per the transistor. ●